

NATHANIEL KONG

+61435218600 | nathanielkong1@gmail.com | github.com/nathanielkong

PROFESSIONAL SUMMARY

Final-year Computer Science student at RMIT University focused on software engineering, artificial intelligence and machine learning. Experience building full-stack applications with React, TypeScript, Python, FastAPI, Java, Spring Boot, PostgreSQL and Docker, including LLM integration and time-series machine learning. Interested in building reliable, customer-focused solutions.

EXPERIENCE

Software Engineer (Contract)

Jan 2026 - Apr 2026

1oak Marketing and Agency Company | Malaysia

Client website designed and developed: [The House of Vows - thehouseofvows.netlify.app](https://thehouseofvows.netlify.app)

- Delivered **2 client websites**, including The House of Vows using Next.js, React and TypeScript; translated client requirements and feedback into responsive pages and enquiry pathways.
- Implemented page metadata, JSON-LD and an XML sitemap; achieved Lighthouse scores of **91/100 SEO, 91/100 accessibility and 100/100 best practices** for The House of Vows.
- Managed Git/GitHub version control, Netlify deployment and post-launch refinements, taking projects from client brief to live website.

PROJECTS

Analyyq - Full-Stack AI Stock Research Platform

React | TypeScript | Python / FastAPI | PostgreSQL | scikit-learn | Gemini API | Docker | GitHub Actions

- Built a full-stack stock research app integrating **3 APIs** (Alpaca, Alpha Vantage and Gemini), with interactive charts, Google/password authentication and persistent chat history.
- Connected LLM prompt interpretation to backend analytics; generated answers from retrieved market data and calculated metrics, caching daily reports across chat sessions to avoid repeated collection and generation.
- Engineered momentum, volatility, volume and benchmark features, with optional VADER news sentiment; trained logistic regression for **1-, 3- and 5-session** direction using gap-aware walk-forward validation.
- In a **6-stock audit**, tuned regularization to lower mean Brier score from **0.2958 to 0.2667**; retained no-edge outputs because the models did not beat naive baselines.
- Built a backend test suite with **203 tests passing in Docker**; containerized the frontend, API and PostgreSQL and configured GitHub Actions for CI, image publishing and deployment.

AI Blockchain Scam Detection Assistant - Final Year Project

Python | REST APIs | OpenAI API | Blockchain APIs

- Developing an AI-assisted platform to analyse cryptocurrency wallets, tokens and transactions for scam-risk indicators.
- Integrating blockchain REST APIs and an LLM to collect risk signals and explain technical indicators in plain language.
- Designing modular backend services to support additional blockchain data providers and detection rules.

EventHub - Full-Stack Event Management Platform

Next.js | React | TypeScript | Java / Spring Boot | REST APIs | PostgreSQL | Docker | JUnit

- Built **6 React/TypeScript components** and the Next.js frontend structure for event discovery and attendee workflows in a **5-person team**.
- Connected discovery and RSVP workflows to **4 Spring Boot REST endpoints**, using typed models, registration-status updates and HTTP error handling.
- Implemented shareable event-detail routes and drag-and-drop image previews with file-type checks and object URL cleanup.
- Ran the frontend and backend in **2 Docker containers**; team CI used GitHub Actions to run JUnit tests before backend image builds.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

Frameworks & Libraries: React, Next.js, FastAPI, Spring Boot, Node.js, SQLAlchemy, JUnit, pytest

AI & Machine Learning: scikit-learn, logistic regression, feature engineering, time-series validation, LLM APIs, VADER

Databases: PostgreSQL | **DevOps:** Docker, Docker Compose, Git, GitHub Actions, CI/CD

Other: REST APIs, HTML/CSS, Figma

EDUCATION

RMIT University | Melbourne, Australia

2023 - 2026

Bachelor of Computer Science

Focus: Artificial Intelligence, Machine Learning & Full-Stack Development

Relevant Coursework: Full Stack Development, Machine Learning, Practical Data Science, Advanced Programming for Data Science, Algorithms & Analysis, C++ Bootcamp

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services | October 2026